Sheaves & Proofs

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Desired background

Some exposure to proof theory (sequent calculus) and semantics (structures, satisfaction, soundness, completeness) for classical predicate logic.

Not required background

- Intuitionistic logic (sequent calculus, Kripke models)
- Basic category theory (categories, functors, natural transformations)
- Basic point-set topology (topologies, covers)

Part 1

Coverage Semantics over Partial Orders

(The logic of localic toposes)

- Kripke semantics
- Cut-elimination via constructive completeness
- Coverage semantics
- The $\neg\neg$-dense coverage.

(Inverted) Kripke semantics

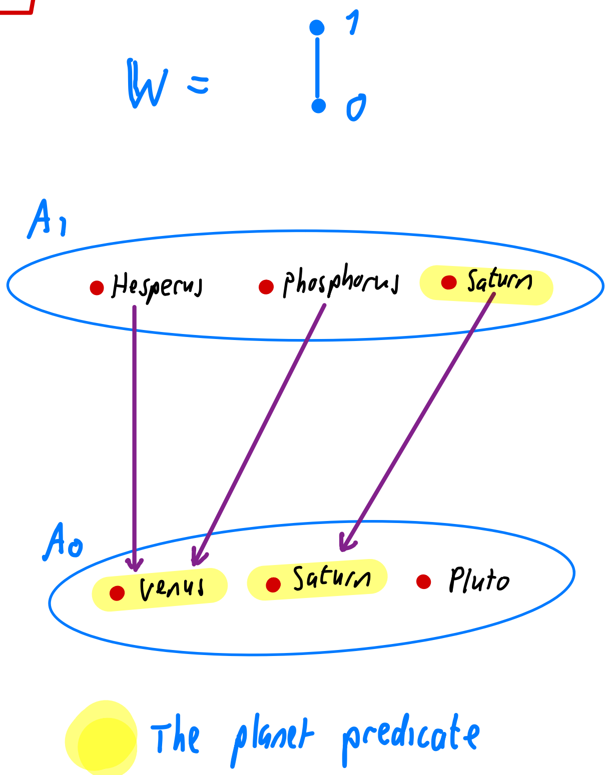
A <u>Kripke model</u> is given by

- A partial order $(W, \leq)$ of <u>worlds</u>
- To every $w \in W$ a set A_w
- <u>Transition functions</u> $(t_{vw} : A_w \to A_v)_{v \leq w}$

  satisfying $t_{uu} = id_{A_u}$; $u \leq v \leq w \Rightarrow t_{uw} = t_{uv} \circ t_{vw}$
- For every predicate P of arity k

  Subsets $P_w \subseteq (A_w)^k$ satisfying

  $v \leq w \Rightarrow t_{vw}(P_w) \subseteq P_v$.

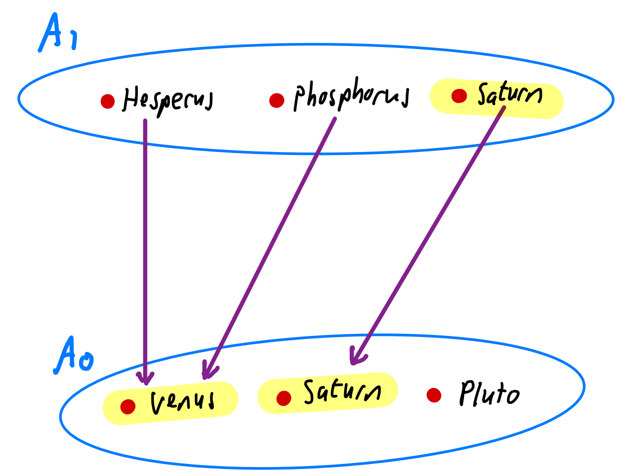


Category-theoretic formulation

A <u>Kripke model</u> is given by

- A partial order $(\mathbb{W}, \leq)$ of <u>worlds</u>
- A functor $A : \mathbb{W}^{op} \to \underline{\mathbf{Set}}$
  (i.e., a <u>presheaf</u> $A \in \mathrm{PSh}(\mathbb{W})$)
- For every predicate P of arity k a <u>subpresheaf</u> $P \subseteq A^k$

$$\mathbb{W} = \begin{tikzcd} 1 \arrow[d, dash] \\ 0 \end{tikzcd}$$



The planet predicate

<u>Notation</u>

$$\begin{aligned} \text{Venus} &= t_{01}(\text{Hesperus}) && \text{(previous slide)} \\ &= A(0<1)(\text{Hesperus}) && \text{(this slide)} \\ &= \text{Hesperus} \cdot 0 && \text{(henceforth)} \end{aligned}$$

Kripke forcing relation

$$w \Vdash \phi$$

$w \Vdash P(a_1,\dots,a_k) \iff (a_1,\dots,a_k) \in P_w$

$w \Vdash a_1 = a_2 \iff a_1 = a_2$

$w \Vdash \phi \wedge \psi \iff w \Vdash \phi$ and $w \Vdash \psi$

$w \Vdash \phi \vee \psi \iff w \Vdash \phi$ or $w \Vdash \psi$

$w \Vdash \bot \iff$ false

$w \Vdash \exists x \phi \iff$ there exists $a \in A_w$ s.t. $w \Vdash \phi[x := a]$

$w \Vdash \phi \to \psi \iff$ for all $v \leq w$ $v \Vdash \phi \cdot v$ implies $v \Vdash \psi \cdot v$

$w \Vdash \forall x \phi \iff$ for all $v \leq w$ and $a \in A_v$ $v \Vdash (\phi \cdot v)[x := a]$

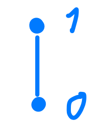
$1 \nVdash \text{Hesperus} = \text{Phosphorus}$

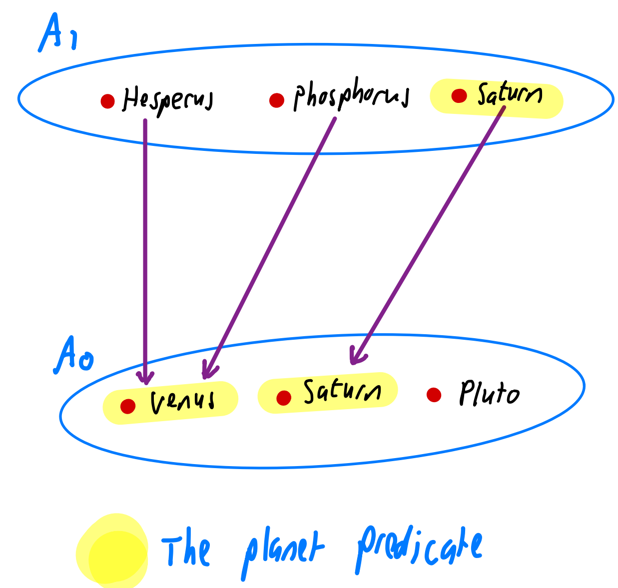
$0 \Vdash \text{Hesperus}\cdot 0 = \text{Phosphorus}\cdot 0$

$1 \nVdash \neg(\text{Hesperus} = \text{Phosphorus})$

$1 \nVdash H = P \vee \neg(H = P)$

$1 \Vdash \neg\neg(H = P)$

$\mathbb{W} =$ 



$\neg\phi := \phi \to \bot$

$w \Vdash \neg\phi \iff \forall v \le w \,.\, v \nVdash \phi\cdot v$

$w \Vdash \neg\neg\phi \iff \forall v \le w \ \neg\neg\ \exists u \le v \ \ u \Vdash \phi\cdot u$

Meta theorems for Kripke semantics

Monotonicity If $w \Vdash \phi$ and $v \le w$ then $v \Vdash \phi$.

Soundness If ϕ is provable in intuitionistic predicate logic then for all Kripke models $(W, \ldots)$ and $w \in W$, $w \Vdash \phi$.

Completeness If, for all Kripke models $(W, \ldots)$ and $w \in W$, $w \Vdash \phi$ then ϕ is provable in intuitionistic predicate logic

Completeness is <u>not</u> provable in a constructive meta-theory.

To achieve constructive completeness, we need more general notion of model.

Coverage base

a.k.a. base for a Grothendieck topology on a poset

A coverage base on a poset $\mathbb{W}$ is a relation $\rhd$ relating pairs $C \rhd w$, where $w \in \mathbb{W}$ and $C \subseteq {\downarrow} w$, satisfying:

$${\downarrow} w := \{ v \in \mathbb{W} \mid v \leq w \}$$

$$\rhd \subseteq \sum_{w \in \mathbb{W}} \mathcal{P}({\downarrow} w)$$

Reflexivity $\{w\} \rhd w$

Transitivity If $C \rhd w$ and, for all $v \in C$, $C_v \rhd v$ then $\bigcup_{v \in C} C_v \rhd w$.

Stability If $C \rhd w$ and $w' \leq w$ then there exists $D \rhd w'$ such that, for all $v' \in D$ there exists $v \in C$ such that $v' \leq v$.

Example coverage bases

① The <u>identity coverage</u> $\{w\} \rhd w$ on any poset $\mathbb{W}$

② For any topological space T, define

$$\mathbb{W} = \mathcal{O}(T) \qquad \text{(open sets)}$$

$$v \leq w \;:\Leftrightarrow\; v \subseteq w$$

$$C \rhd w \;:\Leftrightarrow\; \bigcup C = w \qquad \text{(C covers w)}$$

<u>Exercise</u> Prove that ② satisfies properties of a cover base!

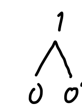
A $\boxed{\underline{\text{separated model}}}$ is given by

- A partial order $(W, \leq)$ of <u>worlds</u> with coverage base $\rhd$
- To every $w \in W$ a set A_w
- <u>Transition functions</u> $(t_{vw} : A_w \to A_v)_{v \leq w}$ s.t. $u \leq v \leq w \Rightarrow t_{uw} = t_{uv} \circ t_{vw}$
- $C \rhd w \Rightarrow \forall a, b \in A_w \ (\forall v \in C,\ t_{vw}(a) = t_{vw}(b)) \Rightarrow a = b$. ($\lhd$-separated)
- For every predicate P of arity k subsets $P_w \subseteq (A_w)^k$ satisfying
  - $\forall \vec{a} \in A_w^k \ (\vec{a} \in P_w \Rightarrow \forall v \leq w,\ t_{vw}(\vec{a}) \in P_v)$
  - $C \rhd w \Rightarrow \forall \vec{a} \in A_w^k \ (\forall v \in C,\ t_{vw}(\vec{a}) \in P_v) \Rightarrow \vec{a} \in P_w$ (P is $\lhd$-closed)

Category-theoretic formulation

A <u>separated model</u> is given by

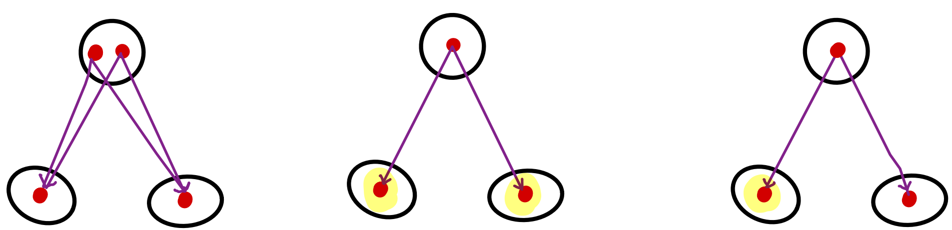
- A partial order $(\mathbb{W}, \leq)$ of worlds
- A <u>$\rhd$-separated</u> presheaf $A : \mathbb{W}^{op} \to \underline{Set}$
- For every predicate P of arity k, a <u>$\rhd$-closed</u> subpresheaf $P \subseteq A^k$

$\mathbb{W} =$ 

$\{0, 0'\} \rhd 1$

$\{0\} \not\rhd 1$

$\{0'\} \not\rhd 1$



not separated   not closed   a separated model

New forcing clauses

$w \Vdash \phi \vee \psi \iff$ there exists a cover $C \rhd w$ such that,
for every $v \in C$, $v \Vdash \phi \cdot v$ or $v \Vdash \psi \cdot v$

$w \Vdash \bot \iff \emptyset \rhd w$

$w \Vdash \exists x\, \phi \iff$ there exists a cover $C \rhd w$ such that,
for every $v \in C$,
there exists $a \in \underline{A}_v$ s.t. $v \Vdash (\phi \cdot v)[x := a]$

N.B.

- The clauses for predicates, $=$, $\wedge$, $\rightarrow$, $\forall$ are as for Kripke semantics

For our example coverage bases

(1) The identity coverage base $\{w\} \rhd w$ on any poset $\mathbb{W}$

Recovers Kripke semantics

(2) The topological cover relation for $\mathbb{W} = \mathcal{O}(T)$.

Recovers topological semantics of intuitionistic logic

Meta theorems for coverage semantics

Monotonicity If $w \Vdash \phi$ and $v \leq w$ then $v \Vdash \phi \cdot v$

Cover property If $C \rhd w$ satisfies, for all $v \in C$, $v \Vdash \phi \cdot v$, then $w \Vdash \phi$.

See warning at the end of Part 2

Soundness If ϕ is provable in intuitionistic predicate logic then for all separated models $(W, \ldots)$ and $w \in W$, $w \Vdash \phi$.

Completeness If, for all separated models $(W, \ldots)$ and $w \in W$, $w \Vdash \phi$ then ϕ is provable in intuitionistic predicate logic

Completeness is now provable in a constructive meta-theory!

Intuitionistic sequent Calculus

Predicate symbols $P(x_1, \dots, x_k)$ $(k \geq 0)$

No function symbols (in particular no constants) (for brevity of exposition)

Equality $t_1 = t_2$

[The only terms are variables, but we use $t, t_1, \dots$ as meta-variables in places where arbitrary terms would be allowed in presence of function symbols.]

Sequent: $\Theta \,;\, \Gamma \Rightarrow \psi$

where Θ is finite set of variables

Γ is finite set of Θ-formulas, ψ is Θ-formula

A Θ-formula is a formula whose free variables are contained in Θ.

Rules

$$\frac{}{\Theta ; \Gamma, P(\vec{t}) \Rightarrow P(\vec{t})}$$

$$\frac{\Theta ; \Gamma, \phi, \phi' \Rightarrow \psi}{\Theta ; \Gamma \ni \phi \wedge \phi' \Rightarrow \psi} \qquad \frac{\Theta ; \Gamma \Rightarrow \psi \quad \Theta ; \Gamma \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \wedge \psi'}$$

$$\frac{\Theta ; \Gamma, \phi \Rightarrow \psi \quad \Theta ; \Gamma, \phi' \Rightarrow \psi}{\Theta ; \Gamma \ni \phi \vee \phi' \Rightarrow \psi} \qquad \frac{\Theta ; \Gamma \Rightarrow \psi}{\Theta ; \Gamma \Rightarrow \psi \vee \psi'} \qquad \frac{\Theta ; \Gamma \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \vee \psi'}$$

$$\frac{}{\Theta ; \Gamma \ni \bot \Rightarrow \psi}$$

$$\frac{\Theta ; \Gamma, \phi' \Rightarrow \psi \quad \Theta ; \Gamma \Rightarrow \phi}{\Theta ; \Gamma \ni \phi \to \phi' \Rightarrow \psi} \qquad \frac{\Theta ; \Gamma, \psi \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \to \psi'}$$

$$\frac{\Theta \uplus x;\ \Gamma, \phi \Rightarrow \psi}{\Theta;\ \Gamma \ni \exists x \phi \Rightarrow \psi} \qquad \frac{\Theta;\ \Gamma \Rightarrow \psi[t/x]}{\Theta;\ \Gamma \Rightarrow \exists x \psi}$$

$$\frac{\Theta;\ \Gamma, \phi[t/x] \Rightarrow \psi}{\Theta;\ \Gamma \ni \forall x\, t \Rightarrow \psi} \qquad \frac{\Theta \uplus x;\ \Gamma \Rightarrow \psi}{\Theta;\ \Gamma \Rightarrow \forall x \psi}$$

$$\frac{\Theta;\ \Gamma \Rightarrow \psi[s/z]}{\Theta;\ \Gamma \ni s=t \Rightarrow \psi[t/z]} \qquad \frac{\Theta;\ \Gamma \Rightarrow \psi[t/z]}{\Theta;\ \Gamma \ni s=t \Rightarrow \psi[s/z]} \qquad \frac{}{\Theta;\ \Gamma \Rightarrow t=t}$$

$$\frac{\Theta;\ \Gamma \Rightarrow \phi \qquad \Theta;\ \Gamma, \phi \Rightarrow \psi}{\Theta;\ \Gamma \Rightarrow \psi}\ \text{Cut}$$

Define $\Theta; \Gamma \vdash \Psi$:$\Leftrightarrow$ the sequent $\Theta; \Gamma \Rightarrow \Psi$ is provable

$\Theta; \Gamma \vdash_{cf} \Psi$:$\Leftrightarrow$ the sequent $\Theta; \Gamma \Rightarrow \Psi$ is cut-free provable.

Lemma (admissible rules with/without cut)

- (identity) $\Theta; \Gamma, \Psi \vdash_{cf} \Psi$
- (weakening) $\Theta; \Gamma \vdash \Psi \implies \Theta, \Theta'; \Gamma, \Gamma' \vdash \Psi$

  $\Theta; \Gamma \vdash_{cf} \Psi \implies \Theta, \Theta'; \Gamma, \Gamma' \vdash_{cf} \Psi$

Proofs Straightforward inductions.

Theorem (Cut admissibility) $\Theta; \Gamma \vdash_{cf} \phi$ and $\Theta; \Gamma, \phi \vdash_{cf} \psi \implies \Theta; \Gamma \vdash_{cf} \psi$

We shall give a semantic proof, by establishing

Soundness $\Theta; \Gamma \vdash \psi \implies \Theta; \Gamma \vDash \psi$

Cut-free completeness $\Theta; \Gamma \vDash \psi \implies \Theta; \Gamma \vdash_{cf} \psi$

where $\vDash$ is semantic entailment w.r.t. separated models.

The proof will be carried out in a constructive meta-theory, so (in principle) it's possible to extract from the proof an algorithm that maps derivations with cuts to cut-free derivations.

Related work: Beth, Veldman et. al., Palmgren, Avigad

Semantic entailment

$\Theta ; \Gamma \vDash \psi$

$:\Longleftrightarrow$

for every poset $\mathbb{W}$ and coverage base $\rhd$ on $\mathbb{W}$, $\quad \forall \mathcal{M}$

for every separated model $(A : \mathbb{W}^{op} \to \underline{\mathrm{set}}, \{\rho_w\}_w)$,

for every $w \in \mathbb{W}$,

for every $\rho : \Theta \to A_w$

if, for every $\phi \in \Gamma$, $\quad w \Vdash \phi[\rho]$, $\quad \Theta ; \Gamma \vDash_{\mathcal{M}} \psi$

then $w \Vdash \psi[\rho]$

.

Proof of soundness: $\Theta; \Gamma \vdash \psi \implies \Theta; \Gamma \vDash \psi$

Consider any $\mathfrak{M}$, (i.e., $\mathbb{W}$, $\triangleright$ and separated model A, ...)

We prove by ind. on derivation of $\Theta; \Gamma \vdash \psi$ that $\Theta; \Gamma \vDash_{\mathfrak{M}} \psi$ holds

Exercise!

Proof of cut-free completeness: $\Theta; \Gamma \vDash \psi \implies \Theta; \Gamma \vdash_{cf} \psi$

In Part 1, we consider just the case of intuitionistic <u>propositional</u> logic

- All predicate symbols have arity 0.
  (Predicates of arity 0 are atomic propositions.)
- No variables, no quantifiers $\forall, \exists$
- No equality $\simeq$

We build a specific separated model $\mathcal{M}_{ctx}$

The modal $\mathcal{M}_{ctx}$ (propositional logic)

$W_{ctx} := \{ \Gamma \mid \Gamma \text{ a finite set of formulas} \}$ (The set of contexts)

$\Gamma' \leq \Gamma \quad :\Leftrightarrow \quad \Gamma' \supseteq \Gamma$

$C \rhd_{ctxr} \Gamma \quad :\Leftrightarrow \quad \dfrac{C}{\Gamma}$ is a derivable context rule

$A_\Gamma := \emptyset$

$() \in P_\Gamma \quad :\Leftrightarrow \quad \Gamma \vdash_{cf} P$. $(\Gamma \Vdash P \quad :\Leftrightarrow \quad \Gamma \vdash_{cf} P)$

Corrected from lecture. (Thanks to Dominik Wehr)

Context rules

$$\frac{\Gamma, \phi, \phi'}{\Gamma \ni \phi \wedge \phi'}$$

$$\frac{\Gamma, \phi \qquad \Gamma, \phi'}{\Gamma \ni \phi \vee \phi'}$$

$$\frac{}{\Gamma \ni \bot}$$

$$\frac{\Gamma, \phi'}{\Gamma \ni \phi \to \phi'} \quad \Gamma \vdash_{cf} \phi$$

For any context Γ and set of contexts C,

$\dfrac{C}{\Gamma}$ is a **derivable context rule** if there exists a derivation tree of contexts with conclusion Γ, in which every leaf is in C.

Lemma Suppose $\dfrac{C}{\Gamma}$ is a derivable context rule.

- (weakening) For any Γ'',

$$\frac{\{\Gamma', \Gamma'' \mid \Gamma' \in C\}}{\Gamma, \Gamma''}$$

is a derivable context rule

The stability of $\rhd$ ctxt follows.

- (Sequenting) For any ψ,

$$\frac{\{\Gamma' \Rightarrow \psi \mid \Gamma' \in C\}}{\Gamma \Rightarrow \psi}$$

is a cut-free derivable sequent rule.

The $\rhd$-closed property of Mctxt follows

Main lemma

$\mathcal{M}_{ctx}$ enjoys the following.

For any $\Gamma \in W_{ctx}$ and formula ϕ

(1) if $\phi \in \Gamma$ then $\Gamma \Vdash \phi$.

(2) if $\Gamma \Vdash \phi$ then $\Gamma \vdash_{cf} \phi$.

Proof of cut-free completeness: $\Gamma \vDash \psi \implies \Gamma \vdash_{cf} \psi$

Suppose $\Gamma \vDash \psi$.

In particular $\Gamma \vDash_{\mathcal{M}_{ctx}, \Gamma} \psi$. $\circledast$

By ① of main lemma, for every $\phi \in \Gamma$, $\Gamma \Vdash \phi$.

By $\circledast$, $\Gamma \Vdash \psi$.

By ② of main lemma, $\Gamma \vdash_{cf} \psi$.

$\square$

Main lemma M_{ctx} enjoys the following.

For any $\Gamma \in W_{ctx}$ and formula ϕ

(1) if $\phi \in \Gamma$ then $\Gamma \Vdash \phi$.

(2) if $\Gamma \Vdash \phi$ then $\Gamma \vdash_{cf} \phi$.

Proof

(1) and (2) are proved simultaneously by induction on ϕ.

$\square$

Case $\phi \equiv \phi_1 \to \phi_2$

(1) $\phi_1 \to \phi_2 \in \Gamma \implies \Gamma \Vdash \phi_1 \to \phi_2$

Suppose $\Gamma' \supseteq \Gamma$ and $\Gamma' \Vdash \phi_1$

By i.h. (2) $\Gamma' \vdash_{cf} \phi_1$

By $\bigcirc$, $\{\Gamma', \phi_2\} \rhd_{ctxt} \Gamma'$

By i.h. (1), $\Gamma', \phi_2 \Vdash \phi_2$

By covering property $\Gamma' \Vdash \phi_2$.

Case $\phi \equiv \phi_1 \to \phi_2$

(2) $\Gamma \Vdash \phi_1 \to \phi_2 \Rightarrow \Gamma \vdash_{cf} \phi_1 \to \phi_2$

By i.h. (1) $\Gamma, \phi_1 \Vdash \phi_1$

By ○, $\Gamma, \phi_1 \Vdash \phi_2$

By i.h. (2) $\Gamma, \phi_1 \vdash_{cf} \phi_2$

Hence $\Gamma \vdash_{cf} \phi_1 \to \phi_2$ by $(\to R)$.

Case $\phi \equiv \phi_1 \vee \phi_2$

(1) $\phi_1 \vee \phi_2 \in \Gamma \Rightarrow \Gamma \Vdash \phi_1 \vee \phi_2$

We have $\{(\Gamma, \phi_1), (\Gamma, \phi_2)\} \rhd_{ctxt} \Gamma$

By i.h. (1), $\Gamma, \phi_1 \Vdash \phi_1$ and $\Gamma, \phi_2 \Vdash \phi_2$.

Hence $\Gamma \vdash \phi_1 \vee \phi_2$.

Case $\phi \equiv \phi_1 \vee \phi_2$

(2) $\Gamma \Vdash \phi_1 \vee \phi_2 \Rightarrow \Gamma \vdash_{cf} \phi_1 \vee \phi_2$

By ○, there is $C \rhd_{ctxt} \Gamma$ such that

for every $\Gamma' \in C$, $\Gamma' \Vdash \phi_1$ or $\Gamma' \Vdash \phi_2$.

So for every $\Gamma' \in C$, $\Gamma' \vdash_{cf} \phi_1$ or $\Gamma' \vdash_{cf} \phi_2$, by i.h (2).

So for every $\Gamma' \in C$, $\Gamma' \vdash_{cf} \phi_1 \vee \phi_2$, by $(\vee R)$ rules.

Since $\frac{C}{\Gamma}$ is admissible, $\Gamma \vdash_{cf} \phi_1 \vee \phi_2$, by sequenting property.

$\square$

Sieve

In a <u>coverage</u> (rather than coverage base) we only allow covers $C \rhd w$, where C is a <u>sieve</u> on w.

A set $C \subseteq W$ is a <u>sieve</u> on $w \in W$ if

- $C \subseteq \downarrow w$
- $C = \downarrow C$ (C is down closed)

If C is a sieve on w and $v \leq w$ then

$$C \cdot v := \{ u \in C \mid u \leq v \}$$

is a sieve on v.

$w \mapsto \mathrm{Sieves}(w)$ defines a presheaf $W^{op} \to \underline{\mathrm{Set}}$

Coverage

A **Coverage** is a relation $C \rhd w$ between $w \in \mathbb{W}$ and sieves C on w satisfying

Reflexivity $\quad \downarrow w \rhd w$

Transitivity $\quad$ If $C \rhd w$ and S is a sieve on w satisfying

$$\forall v \in C, \quad S \cdot v \rhd v$$

then $S \rhd w$.

Stability $\quad$ If $C \rhd w$ and $v \leq w$ then $C \cdot v \rhd v$.

Proposition (Properties of coverages)

Let $\rhd$ be a coverage. Then $\rhd$ satisfies

Monotonicity If $C \rhd w$ and S is a sieve on w then

$$C \subseteq S \Rightarrow S \rhd w$$

Intersection If $C \rhd w$ and $C' \rhd w$ then $C \cap C' \rhd w$

Proof hints For monotonicity use reflexivity + transitivity.

For intersection use transitivity + stability.

Coverages vs. Coverage bases

Proposition

(1) If $\rhd$ is a coverage base then

$$C \rhd w \quad :\Leftrightarrow \quad \exists B \subseteq C,\ B \rhd w \text{ and } C = \downarrow B$$

is a coverage. We say $\rhd$ is the coverage generated by $\rhd$.

See warning at end of part 2

(2) Every coverage $\rhd$ is generated by the coverage base

$$B \rhd w \quad :\Leftrightarrow \quad \downarrow B \rhd w .$$

Proof Exercise!

The $\neg\neg$-dense coverage

On any poset $\mathbb{W}$ define

$$C \rhd_{\neg\neg} w \quad :\Leftrightarrow \quad \text{C is a sieve on } w \text{ and } \forall v \le w \; \neg\neg \, \exists u \le v \; u \in C.$$

Proposition $\rhd_{\neg\neg}$ is a coverage.

Proof Exercise.

$\neg\neg$ Dense $\Rightarrow$ Classical

Proposition Using the coverage $\rhd_{\neg\neg}$, $\forall w \in W$ $w \Vdash \neg\neg\phi \to \phi$

Proof Suppose $w \Vdash \neg\neg\phi$

We show $\{u \leq w \mid u \Vdash \phi\} \rhd_{\neg\neg} w$. Then $w \Vdash \phi$ by cover property.

$\{u \leq w \mid u \Vdash \phi\}$ is a sieve by monotonicity of $\Vdash$

Suppose $v \leq w$. We need to show $\neg\neg \exists u \leq v,\ u \Vdash \phi$.

Suppose for $\perp$ that $\neg \exists u \leq v,\ u \Vdash \phi$.

Then $v \Vdash \neg\phi$.

By $\circ$, $v \Vdash \perp$. So $\emptyset \rhd_{\neg\neg} v$. So $\forall v' \leq v$ $\neg\neg \exists u' \leq v,\ u' \in \emptyset$. I.e. $\perp$.

$\square$

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Part 2

Sheaf Semantics over Categories

(The logic of Grothendieck toposes)

- Presheaf semantics
- Sheaves for coverages
- Coverage semantics over categories
- Extending a separated model to a sheaf model

Let us attempt to directly generalise our cut-free completeness proof from propositional logic to intuitionistic predicate logic

We try to define appropriate $\mathbb{W}_{ctxt}$ and $\triangleright_{ctxt}$

$$\mathbb{W}_{ctxt} := \{\Theta;\Gamma \mid \Theta \text{ a finite set of variables, } \Gamma \text{ a finite set of } \Theta\text{-formulas}\}$$

The set of contexts

$$\Theta';\Gamma' \leq \Theta;\Gamma \;:\Leftrightarrow\; \Theta';\Gamma' \supseteq \Theta;\Gamma$$

$C \triangleright_{ctxt} \Theta;\Gamma \;:\Leftrightarrow\; \dfrac{C}{\Theta;\Gamma}$ is a derivable context rule

Context rules

$$\frac{\Theta ; \Gamma, \phi, \phi'}{\Theta ; \Gamma \ni \phi \wedge \phi'}$$

$$\frac{\Theta ; \Gamma, \phi \qquad \Theta ; \Gamma, \phi'}{\Theta ; \Gamma \ni \phi \vee \phi'}$$

$$\frac{}{\Theta ; \Gamma \ni \bot}$$

$$\frac{\Theta ; \Gamma, \phi'}{\Theta ; \Gamma \ni \phi \to \phi'} \quad \Theta ; \Gamma \vdash_{cf} \phi$$

$$\frac{\Theta \, \forall x ; \Gamma, \phi}{\Theta ; \Gamma \ni \exists x \, \phi}$$

$$\frac{\Theta ; \Gamma, \phi[t/x]}{\Theta ; \Gamma \ni \forall x \, t}$$

However, the weakening property fails for derivable context rules.

E.g. $\dfrac{y \,;\, \exists x\, P(x),\ P(y)}{;\, \exists x\, P(x)}$ is a derivable context rule

but $\dfrac{y \,;\, \exists x\, P(x),\ P(y),\ Q(y)}{y \,;\, \exists x\, P(x),\ Q(y)}$ is not a derivable context rule

Indeed $\mathbb{D}_{ctxt}$ does <u>not</u> satisfy the stability property of a coverage base.

An elegant solution to this problem is to replace the <u>partial order</u> $\mathbb{W}_{ctxt}$ with a <u>category</u>.

Presheaf semantics

A **Presheaf model** is given by

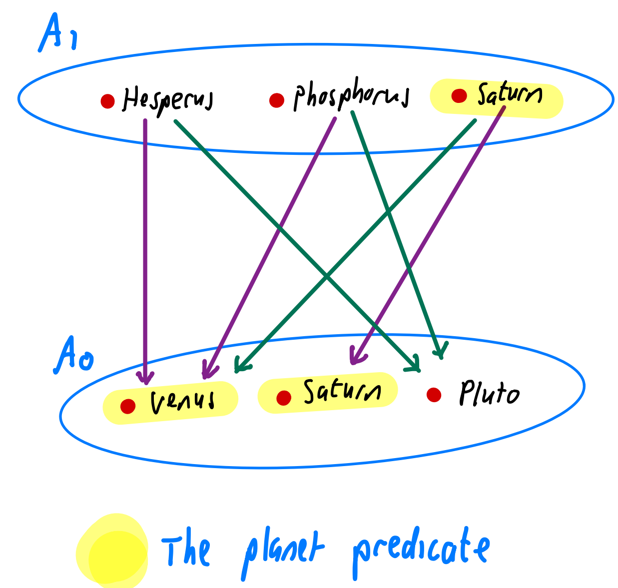
- A (small) category $\mathbb{C}$ of worlds
- To every $x \in \mathbb{C}$ a set A_x
- Transition functions $(t_f : A_x \to A_y)_{y \xrightarrow{f} x}$

  satisfying $z \xrightarrow{g} y \xrightarrow{f} x \Rightarrow t_{f \circ g} = t_g \circ t_f$
- For every predicate P of arity k

  Subsets $P_w \subseteq (A_w)^k$ satisfying

  $y \xrightarrow{f} x \Rightarrow t_f(P_x) \subseteq P_y$

$$\mathbb{C} = \begin{tikzcd} 0 \arrow[r, bend left, "f_0"] \arrow[r, bend right, "f_1"'] & 1 \end{tikzcd}$$

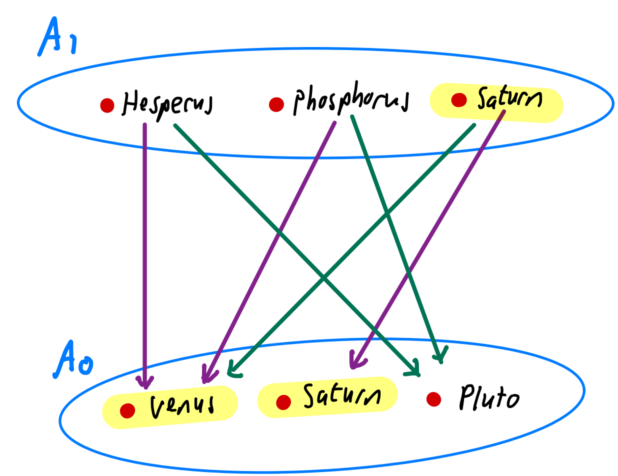


Category-theoretic Formulation

A **Presheaf model** is given by

- A small category $\mathbb{C}$ of worlds
- A functor $A : \mathbb{C}^{op} \to \mathbf{Set}$
  (i.e., a presheaf $A \in \mathrm{PSh}(\mathbb{C})$)
- For every predicate P of arity k a subpresheaf $R \subseteq A^k$

$$\mathbb{C} = \quad 0 \overset{f_0}{\underset{f_1}{\rightrightarrows}} 1$$



The planet predicate

Notation

$\text{Venus} = t_{f_0}(\text{Hesperus})$ (previous slide)

$= A(f_0)(\text{Hesperus})$ (this slide)

$= \text{Hesperus} \cdot f_0$ (henceforth)

Presheaf forcing relation

$$X \Vdash \phi$$

$X \Vdash P(a_1, \ldots, a_k) \iff (a_1, \ldots, a_k) \in P_X$

$X \Vdash a_1 = a_2 \iff a_1 = a_2$

$X \Vdash \phi \wedge \psi \iff X \Vdash \phi$ and $X \Vdash \psi$

$X \Vdash \phi \vee \psi \iff X \Vdash \phi$ or $X \Vdash \psi$

$X \Vdash \bot \iff$ false

$X \Vdash \exists x\, \phi \iff$ there exists $a \in A_X$ s.t. $X \Vdash \phi[x := a]$

$X \Vdash \phi \to \psi \iff$ for all $Y \xrightarrow{f} X$ $\;Y \Vdash \phi \cdot f$ implies $Y \Vdash \psi \cdot f$

$X \Vdash \forall x\, \phi \iff$ for all $Y \xrightarrow{f} X$ and $a \in A_Y$ $\;Y \Vdash (\phi \cdot f)[x := a]$

Meta theorems for Presheaf Semantics

<u>Monotonicity</u> If $X \Vdash \phi$ and $Y \xrightarrow{f} X$ then $Y \Vdash \phi \cdot f$

<u>Soundness</u> If ϕ is provable in intuitionistic predicate logic then for all presheaf models $(\mathbb{C}, \ldots)$ and $X \in \mathbb{C}$, $X \Vdash \phi$.

<u>Completeness</u> If, for all presheaf models $(\mathbb{C}, \ldots)$ and $X \in \mathbb{C}$, $X \Vdash \phi$ then ϕ is provable in intuitionistic predicate logic

(Completeness requires a classical meta-theory.)

Coverage base

a.k.a. base for a Grothendieck topology on a category

A coverage base on a category $\mathbb{C}$ is a relation of the form $C \rhd X$, where $X \in \mathbb{C}$ and $C \subseteq \bigcup_{Y \in \mathbb{C}} \mathbb{C}(Y, X)$

Reflexivity $\quad \{X \xrightarrow{id_X} X\} \rhd X$

Transitivity $\quad$ If $C \rhd X$ and, for all $Y \xrightarrow{f} X \in C$, $C_f \rhd Y$

then $\{Z \xrightarrow{g \circ f} X \mid Y \xrightarrow{f} X \in C,\ Z \xrightarrow{g} Y \in C_f\} \rhd X$

Stability $\quad$ If $C \rhd X$ and $X' \xrightarrow{f} X$ then there exists $D \rhd X'$ such that, for all $Y' \xrightarrow{g'} X' \in D$ there exist $Y \xrightarrow{g} X \in C$ and $Y' \xrightarrow{f'} X'$ such that

$$\begin{tikzcd} Y' \arrow[r, "f'"] \arrow[d, "g'"'] & Y \arrow[d, "g"] \\ X' \arrow[r, "f"'] & X \end{tikzcd}$$

A $\boxed{\underline{\text{separated model}}}$ is given by

- A small category $\mathbb{C}$ of <u>worlds</u> with coverage base $\rhd$
- To every $x \in \mathbb{C}$ a set A_x
- <u>Transition functions</u> $\left((-)\cdot f : A_x \to A_y \right)_{y \xrightarrow{f} x}$ s.t. $(a \cdot f) \cdot g = a \cdot (f \circ g)$
- $C \rhd X \Rightarrow \forall a, b \in A_x \left(\forall y \xrightarrow{f} x \in C,\ a \cdot f = b \cdot f \right) \Rightarrow a = b$. ($\lhd$-separated)
- For every predicate P of arity k subsets $P_w \subseteq (A_w)^k$ satisfying
  - $\forall \vec{a} \in A_x^k \left(\vec{a} \in P_x \Rightarrow \forall y \xrightarrow{f} x,\ \vec{a} \cdot f \in P_y \right)$
  - $C \rhd X \Rightarrow \forall \vec{a} \in A_x^k \left(\forall y \xrightarrow{f} x \in C,\ \vec{a} \cdot f \in P_y \right) \Rightarrow \vec{a} \in P_x$ (P is $\lhd$-closed)

Category-theoretic Formulation

A <u>separated model</u> is given by

- A small category $\mathbb{C}$ of worlds with coverage base $\rhd$
- A <u>$\rhd$-separated</u> presheaf $A : \mathbb{C}^{op} \to \underline{Set}$
- For every predicate P of arity k, a <u>$\rhd$-closed</u> subpresheaf $P \subseteq A^k$

Forcing w.r.t. a coverage

$X \Vdash \phi \vee \psi \iff$

there exists a cover $C \rhd X$ such that,

for every $Y \xrightarrow{f} X \in C$, $Y \Vdash \phi \cdot f$ or $Y \Vdash \psi \cdot f$

$X \Vdash \bot \iff \emptyset \rhd X$

$X \Vdash \exists x\, \phi \iff$ there exists a cover $C \rhd X$ such that,

for every $Y \xrightarrow{f} X \in C$

there exists $a \in \underline{A}_Y$ s.t. $Y \Vdash (\phi \cdot f)[x := a]$

Meta theorems for sheaf semantics

Monotonicity If $X \Vdash \phi$ and $Y \xrightarrow{f} X$ then $Y \Vdash \phi \cdot f$

Cover property If $C \rhd X$ satisfies, for all $Y \xrightarrow{f} X \in C$, $Y \Vdash \phi \cdot f$ then $X \Vdash \phi$.

See warning at end of part 2

Soundness If ϕ is provable in intuitionistic predicate logic then for all separated models $(\mathbb{C}, \rhd, \ldots)$ and $X \in \mathbb{C}$, $X \Vdash \phi$.

Completeness If, for all separated models $(\mathbb{C}, \rhd, \ldots)$ and $X \in \mathbb{C}$, $X \Vdash \phi$ then ϕ is provable in intuitionistic predicate logic

Completeness has an elegant proof in a constructive meta-theory

To prove completeness we use a category based model

The category $\mathbb{C}_{ctx}$

Objects: $\Theta ; \Gamma$ (left-hand sides of sequents)

Morphisms: $\sigma : \Theta' ; \Gamma' \to \Theta ; \Gamma$

$\sigma : \Theta \to \Theta'$ is an injective function

and $\Gamma[\sigma] \subseteq \Gamma'$

The coverage base $\rhd_{ctx}$

$C \rhd \Theta ; \Gamma$ $:\Leftrightarrow$ $\dfrac{C}{\Theta ; \Gamma}$ is a derivable context rule.

$\rhd_{ctx}$ is a coverage base

Example illustrating stability

$$
\begin{tikzcd}
 & y \,;\, \exists x\, P(x),\, P(y) \arrow[d, phantom, "\bigtriangledown"] \\
y \,;\, \exists x\, P(x),\, Q(y) \arrow[r, "\substack{\longmapsfrom \\ !}"'] & ;\, \exists x\, P(x)
\end{tikzcd}
$$

$\rhd_{ctxt}$ is a coverage base

Example illustrating stability

$$\begin{tikzcd}
y, z \,;\, \exists x\, P(x), Q(y), P(z) \arrow[r, "z \mapsfrom y"] \arrow[d, phantom, "y \mapsfrom y"] & y \,;\, \exists x\, P(x), P(y) \arrow[d, phantom, "!"] \\
y \,;\, \exists x\, P(x), Q(y) \arrow[r, "\mapsfrom\ !"'] & ;\, \exists x\, P(x)
\end{tikzcd}$$

Definition of full separated model $\mathfrak{M}_{ctx}$

- Category of worlds $\mathbb{C}_{ctx}$
- Coverage base $\rhd_{ctx}$
- Separated presheaf

$$A_{\Theta;\Gamma} := \{ t \mid vars(t) \subseteq \Theta \} / \sim_{\Theta;\Gamma}$$

$$t \sim_{\Theta;\Gamma} t' :\Leftrightarrow \forall\ \Theta';\Gamma' \xrightarrow{\sigma} \Theta;\Gamma \text{ and } \Theta' \uplus [z]\text{-formula } \psi[z]$$
$$\Theta';\Gamma' \vdash_{cf} \psi[t[\sigma]] \Leftrightarrow \Theta';\Gamma' \vdash_{cf} \psi[t'[\sigma]]$$

$$[t] \cdot \sigma := [t[\sigma]]$$

- Interpretation of predicates

$$([t_1], \dots, [t_n]) \in P_{\Theta;\Gamma} :\Leftrightarrow \Theta;\Gamma \vdash_{cf} P(t_1, \dots, t_n)$$

The proof of cut-free completeness

entailment w.r.t. category-based separated models.

$$\Theta ; \Gamma \vDash \psi \quad \Rightarrow \quad \Theta ; \Gamma \vdash_{cf} \psi$$

now proceeds along similar lines to the propositional case

Once again, the main step is to prove

Main lemma $\mathcal{N}_{ctx}$ enjoys the following.

For any $\Theta ; \Gamma \in \mathbb{C}_{ctx}$ and Θ-formula ϕ

(1) if $\phi \in \Gamma$ then $\Theta ; \Gamma \Vdash \phi[\vec{x} := \overrightarrow{[x]}]$

(2) if $\Theta ; \Gamma \Vdash \phi[\vec{x} := \overrightarrow{[x]}]$ then $\Theta ; \Gamma \vdash_{cf} \phi$.

We have obtained completeness for intuitionistic predicate logic with respect to category-based separated models.

$$\left(\forall \mathcal{N} = (\mathbb{C}, \triangleright, \dots),\ \Theta; \Gamma \vDash_{\mathcal{N}} \psi\right) \Rightarrow \Theta; \Gamma \vdash \psi$$

This can be strengthened to completeness with respect to poset-based separated models

$$\left(\forall \mathcal{M} = (\mathbb{W}, \triangleright, \dots),\ \Theta; \Gamma \vDash_{\mathcal{M}} \psi\right) \Rightarrow \Theta; \Gamma \vdash \psi$$

using the Diaconescu cover, which can be used to construct a poset-based model out of a category-based model, preserving logical properties.

Sheaf models

Separated Model

- A small category $\mathbb{C}$ of worlds with coverage base $\rhd$
- A separated presheaf $A : \mathbb{C}^{op} \to \underline{Set}$
- For every predicate P of arity k, a closed subpresheaf $P \subseteq A^k$

Sheaf model (a special case of separated model)

- A small category $\mathbb{C}$ of worlds with coverage base $\rhd$
- A sheaf $A : \mathbb{C}^{op} \to \underline{Set}$ (every sheaf is a fortiori separated)
- For every predicate P of arity k, a closed subpresheaf $P \subseteq A^k$ $\equiv$ subsheaf

Sheaf on a topological space T

Given a presheaf $A : \mathcal{O}(T)^{op} \to \underline{\mathbf{Set}}$

- A <u>matching family</u> for a cover $C \rhd W$ is a family $(a_v \in A_v)_{v \in C}$ such that $\forall u, v \in C \quad a_u \cdot (u \cap v) = a_v \cdot (u \cap v)$
- An <u>amalgamation</u> for a matching family $(a_v \in A_v)_{v \in C}$ is an element $a_w \in A_w$ s.t. $\forall v \in C \quad a_v = a_w \cdot v$.
- The presheaf A is a <u>sheaf</u> if every matching family has a unique amalgamation. (Separated $\Leftrightarrow$ at most one amalgamation!)

<u>Example sheaf</u> $C : \mathcal{O}(T) \to \underline{\mathbf{Set}} \qquad C(u) := \{ f : u \to \mathbb{R} \mid f \text{ continuous} \}$

$f \cdot v := f|_v \quad$ for $f \in C(u)$ and $v \subseteq u$

Sheaf for a coverage base (over a poset)

Given a presheaf $A : W^{op} \to \underline{\mathrm{Set}}$ and coverage base $\rhd$ on W.

- A <u>matching family</u> for a cover $C \rhd w$ is a family $(a_v \in A_v)_{v \in C}$ such that $\forall v, v' \in C$ $\forall u \leq v, v'$, $a_v \cdot u = a_{v'} \cdot u$

- An <u>amalgamation</u> for a matching family $(a_v \in A_v)_{v \in C}$ is an element $a_w \in A_w$ s.t. $\forall v \in C$ $a_v = a_w \cdot v$.

- The presheaf A is a <u>sheaf</u> if every matching family has a unique amalgamation. (Separated $\Leftrightarrow$ at most one amalgamation!)

Sheaf for a coverage base on a category

Given a presheaf $A : \mathbb{C}^{op} \to \underline{\mathrm{Set}}$ and coverage $\rhd$ on $\mathbb{C}$.

- A <u>matching family</u> for a cover $C \rhd X$ is a family $(a_f \in A_Y)_{Y \xrightarrow{f} X \in C}$ such that $\forall\, Y \xrightarrow{f} X,\ Y' \xrightarrow{f'} X \in C$, $\forall$
$$\begin{tikzcd} Z \arrow[r, "g"] \arrow[d, "g'"'] & Y \arrow[d, "f"] \\ Y' \arrow[r, "f'"'] & X \end{tikzcd}$$
in $\mathbb{C}$, $a_f \cdot g = a_{f'} \cdot g'$.

- An <u>amalgamation</u> for a matching family $(a_f \in A_Y)_{Y \xrightarrow{f} X \in C}$ is an element $a \in A_X$ s.t. $\forall\, Y \xrightarrow{f} X \in C$, $a_f = a \cdot f$.

- The presheaf A is a <u>sheaf</u> if every matching family has a unique amalgamation. (Separated $\Leftrightarrow$ at most one amalgamation!)

There is no loss of generality in considering sheaf models

Theorem

Let $(\mathbb{C}, \rhd)$ be a site (small category + coverage (base)).

Every separated model $\mathcal{N}$ over $(\mathbb{C}, \rhd)$ has a dense elementary embedding into a sheaf model $\mathcal{N}^+$ over $(\mathbb{C}, \rhd)$.

$$\mathcal{N} \hookrightarrow \mathcal{N}^+$$

Moreover, this characterises $\mathcal{N}^+$ up to isomorphism.

Given presheaves $A, B : \mathbb{C}^{op} \to \underline{\mathrm{Set}}$

A presheaf map $\alpha : A \to B$ is a family of functions

$$\left(\alpha_X : A_X \to B_X \right)_{X \in \mathbb{C}}$$

such that $\forall Y \xrightarrow{f} X$ in $\mathbb{C}$ and $a \in A_X$, $\alpha_Y(a \cdot f) = (\alpha_X(a)) \cdot f$.

I.e., a presheaf map is a natural transformation $\alpha : A \Rightarrow B$.

α is an embedding of presheaves if every α_X is injective.

α is dense w.r.t. $\rhd$ if

$$\forall b \in B_X \; \exists C \rhd X \text{ s.t. } \forall Y \xrightarrow{f} X \in C, \; \exists a \in A_Y \text{ s.t. } \alpha_Y(b) = a \cdot f.$$

Given separated models $\mathcal{N} = (A, (P)_P)$ and $\mathcal{N}' = (A', (P')_P)$ an **embedding of models** is a presheaf embedding $\alpha : A \to A'$ that further satisfies

For all $X \in \mathbb{C}$ and $a_1, \dots, a_k \in A_X$

$$(a_1, \dots, a_k) \in P_X \iff (\alpha_X(a_1), \dots, \alpha_X(a_k)) \in P'_X$$

It is an **elementary embedding** if, for all formulas ϕ,

$$X \Vdash^{\mathcal{N}} \phi(a_1, \dots, a_k) \iff X \Vdash^{\mathcal{N}'} \phi(\alpha_X(a_1), \dots, \alpha_X(a_k))$$

Grothendieck's $(\cdot)^+$ construction

Let $\mathbb{C}$ be a small category with coverage base $\rhd$.

Given a presheaf $A : \mathbb{C} \to \underline{\mathrm{Set}}$ define $A^+ . \mathbb{C} \to \underline{\mathrm{Set}}$

$$A^+_X := \{ (C, (a_f)_{f \in C}) \mid C \rhd X \text{ , } (a_f) \text{ is a matching family for } C \} / \sim$$

$$(C, (a_f)_f) \sim (C', (a'_{f'})_{f'}) \iff \text{for every commuting square in } \mathbb{C}$$

$$\begin{tikzcd} \bullet \arrow[r, "g"] \arrow[d, "g'"'] & \bullet \arrow[d, "f \in C"] \\ \bullet \arrow[r, "f' \in C'"'] & X \end{tikzcd} \quad , \quad a_f \cdot g = a'_{f'} \cdot g'$$

Theorem (Grothendieck)

- A^+ is separated, for any presheaf A.
- If A is separated, then A^+ is a sheaf, and $a \in A_x \mapsto [(\{id_x\}, \{a\})]$. $A \to A^+$ is a dense embedding of presheaves.

Extending a model

Theorem

Suppose $\alpha : A \Rightarrow A'$ is a dense embedding between separated presheaves.
Let $\mathcal{N} = (A, (P)_P)$ be any separated model.
For any k-ary predicate P, define $P' \subseteq (A')^k$ by

$$(a'_1, \dots, a'_n) \in P'_X :\Leftrightarrow \exists C \rhd X \text{ s.t. } \forall Y \xrightarrow{f} X \in C$$
$$\exists a_1, \dots, a_n \in A_Y . \ \alpha_Y(a_1) = a'_1 \cdot f, \ \dots, \alpha_Y(a_n) = a'_n \cdot f$$
$$\text{and } (a_1, \dots, a_n) \in P_Y$$

Then $\mathcal{N}' := (A', (P')_P)$ is a separated model and $\alpha : \mathcal{N} \Rightarrow \mathcal{N}'$ is an elementary embedding.

Proof

We just look at the elementary embedding property.

We prove

$$X \Vdash^{N} \phi(a_1, \dots, a_k) \iff X \Vdash^{N'} \phi(\alpha_X(a_1), \dots, \alpha_X(a_k))$$

by induction on ϕ.

We consider just one case.

Case $\phi \equiv \exists x\, \phi'$

$\Rightarrow$ Suppose $X \Vdash (\exists x\, \phi')[a_1, \ldots, a_n]$

There exists $C \rhd X$ s.t. $\forall\, Y \xrightarrow{f} X \in C$, $\exists\, a \in X_Y$ s.t.

$$Y \Vdash^N \phi'[a_1 \cdot f, \ldots, a_n \cdot f, a]$$

By i.h., $\quad Y \Vdash^{N'} \phi'[\alpha_Y(a_1 \cdot f), \ldots, \alpha_Y(a_k \cdot f), \alpha_Y(a)]$

i.e., $\quad Y \Vdash^{N'} \phi'[\alpha_X(a_1) \cdot f, \ldots, \alpha_X(a_k) \cdot f, \alpha_Y(a)]$

Hence $\quad X \Vdash^{N'} (\exists x\, \phi')[\alpha_X(a_1), \ldots, \alpha_X(a_n)]$.

$\Leftarrow$ Suppose $X \Vdash^{N'} (\exists x\, \phi')[\alpha_X(a_1), \ldots, \alpha_X(a_k)]$

There exists $C \rhd X$ s.t. $\forall Y_f \xrightarrow{f} X$, $\exists a_f' \in A'_{Y_f}$ s.t.

$$Y_f \Vdash^{N'} \phi'[\alpha_X(a_1)\cdot f, \ldots, \alpha_X(a_k)\cdot f, a_f']$$

By density $\exists C_f \rhd Y_f$ $\forall Z \xrightarrow{g} Y_f \in C_f$ $\exists a_g \in A_Z$ $\alpha_Z(a_g) = a_f' \cdot g$.

By transitivity of $\rhd$, $D := \{ f \circ g \mid f \in C, g \in C_f \} \rhd X$ warning!

Moreover, $\forall Z \xrightarrow{f \circ g} X \in D$, $Z \Vdash^{N'} \phi'[\alpha_Z(a_1 \cdot f \cdot g), \ldots, \alpha_Z(a_k \cdot f \cdot g), \alpha_Z(a_g)]$.

By i.h., $Z \Vdash^{N} \phi'[a_1 \cdot f \cdot g, \ldots, a_k \cdot f \cdot g, a_g]$

Hence, $Z \Vdash^N (\exists x\, \phi')[a_1, \ldots, a_k]$.

$\square$

The $\boxed{\text{warning}}$ is that $\{f \circ g \mid f \in C, g \in C_f\}$ is not a good definition, as it uses choice to select a chosen C_f for every f.

(Similar issues arise in proving that $\Vdash$ enjoys the cover property in the case of $\vee$ and $\exists$, and in proving transitivity for the coverage generated by a coverage base.)

Two possible solutions

- Define $\Vdash$ w.r.t. a coverage rather than coverage base.
- Or work with coverage bases formulated using indexed families

$$\{Y_i \xrightarrow{f_i} X\}_{i \in I} \rhd X .$$

Perspective

Coverage $\equiv$ Grothendieck topology

Coverage base $\equiv$ base for a Groth. top. (one sees other variant definitions)

The category of presheaf maps between sheaves is a Groth. topos.

All Groth. toposes arise in this way.

A sheaf model over $(\mathbb{C}, \rhd)$ is just

- An object $A \in \mathcal{E} := \underline{Sh}(\mathbb{C}, \rhd)$
- For every predicate P a subobject $P \rightarrowtail A^k$ in $\mathcal{E}$

That is, it is an <u>ordinary relational structure</u> from the perspective of $\mathcal{E}$.

The forcing relation captures truth in the internal logic of $\mathcal{E}$. E.g.,

$$\forall x \in \mathbb{C},\ x \Vdash \phi \iff \mathcal{E} \text{ validates } \mathbb{M} \vDash \phi$$

Internal Tarskian semantics

Literature (will be expanded in due course.)

The literature I know on sheaf semantics approaches it via first understanding toposes and their logic, requiring substantial category theory.

My favourite book that takes this approach is

- Mac Lane & Moerdijk *Sheaves in Geometry and Logic*

A more gently paced presentation (with less content) can be found in

- Goldblatt *Topoi*